Chapter 1

Minimization Using Polynomials

In this chapter we will study *combinatorial discrepancy*, which is the following basic combinatorial and algorithmic problem.

> Let X be a finite set, and let $\mathcal{R}$ be a set of subsets of X. We call $(X, \mathcal{R})$ a *set system* with *vertex set* X.
>
> Given $(X, \mathcal{R})$, a *two-coloring* of X is an assignment of a -1 or a $+1$ color to each element of X. Let χ denote a fixed two-coloring of X. That is,
>
> $$\chi \colon X \to \{-1, +1\}\,.$$
>
> Then, the *discrepancy* of a set $S \subseteq X$ with respect to χ is defined as
>
> $$\operatorname{disc}_\chi(S) = \left| \sum_{v \in S} \chi(v) \right|.$$
>
> Define the *discrepancy* of $\mathcal{R}$ with respect to χ as
>
> $$\operatorname{disc}_\chi(\mathcal{R}) = \max_{S \in \mathcal{R}} \operatorname{disc}_\chi(S).$$

Our goal is to find a two-coloring χ that minimizes $\operatorname{disc}_\chi(\mathcal{R})$.

> **Remark:** It is easy to find a coloring that minimizes the discrepancy of a *single* set S: just color half of the vertices in S with -1, and the other half with $+1$. Then, $\operatorname{disc}_\chi(S) = 0$ if $|S|$ is even, and $\operatorname{disc}_\chi(S) = 1$ if $|S|$ is odd.
>
> The challenge lies in finding a single coloring such that *all* sets of $\mathcal{R}$ have low discrepancy with respect to it. This is much more difficult, because the different constraints may conflict with each other.

A simple probabilistic argument (Exercise 1.1.5 (page 12)) shows that we can achieve small discrepancy as a function of $|X|$ and $|\mathcal{R}|$.

> **Lemma 1.1.** *Let $(X, \mathcal{R})$ be a finite set system, and let $n = |X|$ and $m = |\mathcal{R}| \geq 2$. Let χ be a random two-coloring of X, where, independently, each $v \in X$ is assigned one of the two*

colors with probability $1/2$*. Then, with probability at least* $1 - 1/m$*,*

$$\mathrm{disc}_\chi(\mathcal{R}) = O\left(\sqrt{n \ln m}\right).$$

Lemma 1.1 (page 5) gives, in fact, *two* things:

1. there is a coloring of X with discrepancy $O\left(\sqrt{n \ln m}\right)$, and
2. there is a simple randomized algorithm that computes such a coloring with probability at least $1 - 1/m$.

Our objective in this chapter is to construct such a coloring via a *deterministic* algorithm.

> **Remark:** Viewed combinatorially, Lemma 1.1 (page 5) implies that out of all the 2^n possible two-colorings of X, a large fraction—at least $(1 - 1/m)\, 2^n$—have discrepancy $O\left(\sqrt{n \ln m}\right)$.
>
> But can we find *any one* such two-coloring by an efficient deterministic algorithm?

In the first section of this chapter, we analyze an iterative algorithm that follows a simple greedy strategy. Given $X = \{v_1, v_2, \ldots, v_n\}$, we will assign the colors to the elements of X one by one, as follows.

> For $t = 1, \ldots, n$:
>
> — color the vertex v_t to minimize the discrepancy of $\mathcal{R}$ (so far).
>
> That is, among all the sets of $\mathcal{R}$ containing v_t, let $S \in \mathcal{R}$ be a set with the **maximum** discrepancy with respect to the colors assigned to $\{v_1, \ldots, v_{t-1}\}$.
>
> Assign v_t the color that *decreases* the discrepancy of S. If the discrepancy of S is 0, assign v_t the color $+1$.

The first section shows that this fails, quite badly. The reason is linked, in a fundamental way, to the use of the **maximum** function above:

> if we make each coloring decision by focusing only on the set with the maximum discrepancy, then the discrepancy of many other sets can increase over time.
>
> Thus, at the end, there will be at least one set with too high a discrepancy.

This problem is really the *raison d'être* of this text—algorithms guided by the natural objective function involving maxima or minima often do not work for this reason.

The second section is devoted to the main new insight:

> *instead of minimizing the maximum discrepancy at each iteration, we will consider a more global criterion that takes into account the discrepancy of* ***all*** *the sets of* $\mathcal{R}$*.*
>
> Not surprisingly, in this global criterion, sets with high discrepancy will be given more importance.
>
> And then we can apply the greedy algorithm—but this time being greedy according to the new criterion.

We will prove that this works out—in fact, we will recover the asymptotic bound of Lemma 1.1 (page 5)!

1.1 When Minimizing the Maximum Fails (the obvious algorithm fails)

> *So many vows . . . they make you swear and swear. Defend the king. Obey the king. Keep his secrets. Do his bidding. Your life for his. But obey your father. Love your sister. Protect the innocent. Defend the weak. Respect the gods. Obey the laws. It's too much. No matter what you do, you're forsaking one vow or the other.*
>
> A Clash of Kings (George R. R. Martin)

The main result of this section is the following.

Theorem 1.2. *Given a finite set system $(X, \mathcal{R})$ with $n = |X| \geq 2$ even, consider the following greedy algorithm to compute a two-coloring $\chi\colon X \to \{-1, +1\}$ with low discrepancy.*

Greedy Discrepancy Algorithm $\left(X = \{v_1, \ldots, v_n\}, \mathcal{R}\right)$

For $t = 1, \ldots, n$:

Step 1. *Among all the sets of $\mathcal{R}$ containing v_t (we assume that every element of X lies in at least one set of $\mathcal{R}$), let S^t be a set with maximum discrepancy:*

$$S^t = \arg \max_{\substack{S \in \mathcal{R}: \\ v_t \in S}} \operatorname{disc}^t(S),$$

where $\operatorname{disc}^t(S)$ is the discrepancy of S at the start of iteration t, with respect to the colors of $\{v_1, \ldots, v_{t-1}\}$. That is,

$$\operatorname{disc}^t(S) = \left| \sum_{\substack{v_i \in S: \\ i<t}} \chi(v_i) \right|.$$

Step 1. *If $\operatorname{disc}^t(S^t) > 0$, set $\chi(v_t)$ to be the color that* decreases *the discrepancy of S^t.*
If $\operatorname{disc}^t(S^t) = 0$, we set $\chi(v_t) = +1$.

*There exists a set system $(X, \mathcal{R})$ with $n = |X| = |\mathcal{R}|$ even, $n \geq 2$, an ordering of its elements, and an adversarial tie-breaking sequence (for **Step 1.**) such that the coloring constructed by the algorithm has discrepancy at least $n/2$.*

Remark: Since the algorithm does not prescribe a tiebreaking rule, our adversarial construction fixes a particular tiebreaking sequence (the worst case for the algorithm), showing that without a careful tiebreaking policy, the greedy algorithm can perform arbitrarily poorly.

Overview of ideas. The proof is via an adversary argument, where we *incrementally* construct the required set system $\mathcal{R}$ over the n iterations. This is possible since ***Greedy Discrepancy Algorithm***, during the t-th iteration, completely ignores the vertices $v_{t+1}, \ldots, v_n$. Thus, we do not need to have the set system fully constructed at the very start—only the assignments of $\{v_1, \ldots, v_t\}$ to sets of $\mathcal{R}$ need to be fixed at the start of the t-th iteration.

The key idea is that on receiving the element $v_t \in X$ at the start of the t-th iteration, $t \geq 2$, we (the adversary) will add v_t to the sets of $\mathcal{R}$ such that coloring v_t so as to reduce the discrepancy of the

maximum-discrepancy set containing v_t will, in fact, increase the discrepancy of a lot of other sets of $\mathcal{R}$.

> In particular, in every iteration after the first, the discrepancy of half the sets will increase, while the discrepancy of only one set actually decreases!

Thus, at the end of the algorithm, by the pigeonhole principle, there will be a set with large discrepancy.



Proof of Theorem 1.2 (page 7). Set $k = n/2$.

We adaptively construct the sets as the algorithm proceeds with the n iterations. The set system $\mathcal{R}$ consists of the following $n = 2k$ sets:

$$P_1, \ldots, P_k, \quad N_1, \ldots, N_k.$$

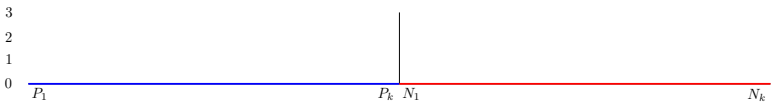
Initially, all these sets are empty, and we allocate elements to them in the order $v_1, \ldots, v_n$. At all times, we maintain the following invariant:

> the sets $P_1, \ldots, P_k$ have signed (that is, just the sum of the color values) discrepancy ≥ 0, and the sets $N_1, \ldots, N_k$ have signed discrepancy ≤ 0.

We now describe the details of the adversary argument. Note that in case of ties, the adversary will make a tie-breaking choice.

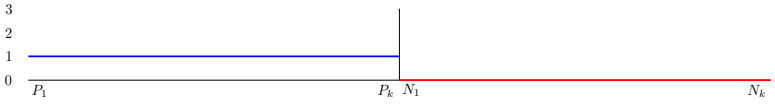
The figures below give a visual representation of the discrepancy of the sets of $\mathcal{R}$ after v_t has been assigned a color.

Initial situation : all $2k$ sets have discrepancy 0.



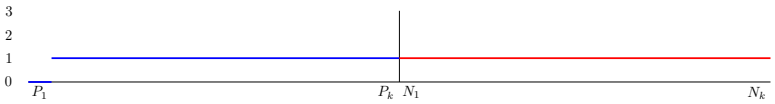
On the arrival of v_1 : add v_1 to $P_1, \ldots, P_k$.

> ***Greedy Discrepancy Algorithm*** sets $\chi(v_1) = +1$ (since all sets containing v_1 have discrepancy 0).



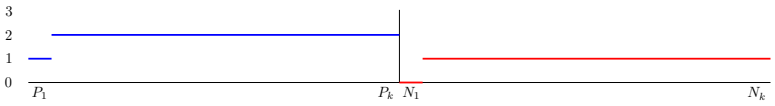
On the arrival of v_2 : add v_2 to P_1 and $N_1, \ldots, N_k$.

> To minimize $\mathrm{disc}^2(P_1)$, ***Greedy Discrepancy Algorithm*** sets $\chi(v_2) = -1$.



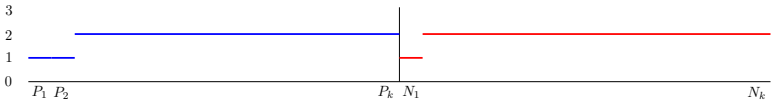
On the arrival of v_3 : add v_3 to N_1 and $P_1, \ldots, P_k$.

To minimize $\mathrm{disc}^3(N_1)$, ***Greedy Discrepancy Algorithm*** sets $\chi(v_3) = +1$ (note that there are several choices for S^3; for our argument, we assume that ***Greedy Discrepancy Algorithm*** chooses N_1).



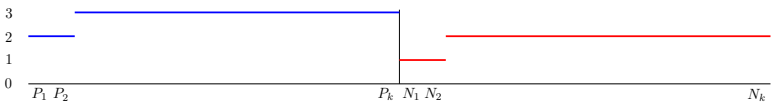
On the arrival of v_4 : add v_4 to P_2 and $N_1, \ldots, N_k$.

To minimize $\mathrm{disc}^4(P_2)$, ***Greedy Discrepancy Algorithm*** sets $\chi(v_4) = -1$.



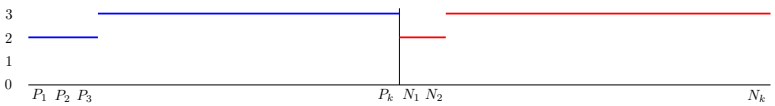
On the arrival of v_5 : add v_5 to N_2 and $P_1, \ldots, P_k$.

To minimize $\mathrm{disc}^5(N_2)$, ***Greedy Discrepancy Algorithm*** sets $\chi(v_5) = +1$.



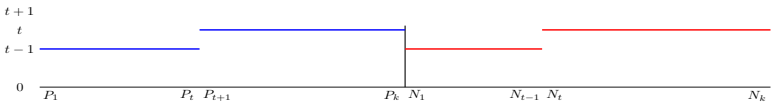
On the arrival of v_6 : add v_6 to P_3 and $N_1, \ldots, N_k$.

To minimize $\mathrm{disc}^6(P_3)$, ***Greedy Discrepancy Algorithm*** sets $\chi(v_6) = -1$.



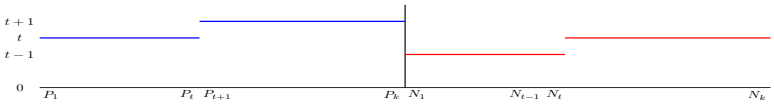
$\vdots$

Situation before iteration $2t+1$: We have $\mathrm{disc}^{2t+1}(P_1) = \cdots = \mathrm{disc}^{2t+1}(P_t) = t-1$, $\mathrm{disc}^{2t+1}(P_{t+1}) = \cdots = \mathrm{disc}^{2t+1}(P_k) = t$, and $\mathrm{disc}^{2t+1}(N_1) = \cdots = \mathrm{disc}^{2t+1}(N_{t-1}) = t-1$, $\mathrm{disc}^{2t+1}(N_t) = \cdots = \mathrm{disc}^{2t+1}(N_k) = t$.



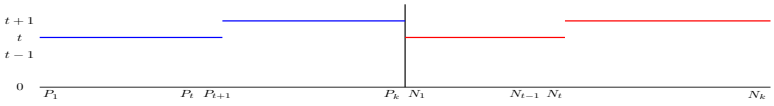
On the arrival of v_{2t+1} : add v_{2t+1} to N_t and $P_1, \ldots, P_k$.

To minimize $\mathrm{disc}^{2t+1}(N_t)$, ***Greedy Discrepancy Algorithm*** sets $\chi(v_{2t+1}) = +1$.



On the arrival of v_{2t+2} : add v_{2t+2} to P_{t+1} and $N_1, \ldots, N_k$.

To minimize $\mathrm{disc}^{2t+2}(P_{t+1})$, ***Greedy Discrepancy Algorithm*** sets $\chi(v_{2t+2}) = -1$.



$\vdots$

The process ends after iteration $n = 2k$. Then, we have $\mathrm{disc}^{2k+1}(P_1) = \cdots = \mathrm{disc}^{2k+1}(P_k) = k - 1$ and $\mathrm{disc}^{2k+1}(N_1) = \cdots = \mathrm{disc}^{2k+1}(N_{k-1}) = k - 1$, $\mathrm{disc}^{2k+1}(N_k) = k$.

Thus at the end, we will have

$$\mathrm{disc}(N_k) = \frac{n}{2},$$

as claimed. □

Remark: We can generalize this construction for an arbitrary number m, $m \geq n$, of sets.

For this, we simply add an additional $\lceil \log m \rceil$ elements to X, and we create distinct subsets of these elements to obtain the required number of different sets in $\mathcal{R}$.

This operation can only change the discrepancy by an additive term of $\lceil \log m \rceil$.

Remark: It is useful to see the above construction by assigning, and tracking, suitable *weights*:

1. assign, to each $S \in \mathcal{R}$, the weight
$$y_S = \mathrm{disc}\,(S)\,.$$
2. construct sets so that by following ***Greedy Discrepancy Algorithm***, after the t-th iteration, we have
$$\sum_S y_S = \Omega\,(t \cdot k)\,.$$

More precisely, at the end of the algorithm, we have

$$\sum_S y_S = k(k-1) + (k-1)^2 + k = 2k^2 - 2k + 1.$$

3. at the end, by the pigeonhole principle, one set of $\mathcal{R}$ must have weight at least $\frac{1}{2k}$-th of the total weight. That is, one set of $\mathcal{R}$ must have discrepancy at least

$$\frac{2k^2 - 2k + 1}{2k} = k - 1 + \frac{1}{2k}.$$

As discrepancy is an integer, this set must have discrepancy at least $k = n/2$.



Bibliography and discussion. This example was constructed by the authors for pedagogical purposes. Constructions similar in spirit are known for other problems.
We refer the reader to the beautiful books by Chazelle [Cha00] and Matoušek [Mat99] for more information on discrepancy.

EXERCISES

EXERCISES STILL UNDER CONSTRUCTION

1. Generalize Theorem 1.2 (page 7) so that the counterexample contains an arbitrary number $m \geq n$ of sets.

   Solution. We can generalize this construction to an arbitrary number $m \geq n$ of sets (if $m = n$ there is nothing to add). Add $L = \lceil \log_2 m \rceil$ new elements $v_{n+1}, \ldots, v_{n+L}$ to X, and add $m - n$ further sets, chosen as distinct subsets of these new elements so that every new element lies in at least one of them (possible, since $2^L \geq m$). The new sets contain none of the original vertices, so the first n iterations proceed exactly as above, and N_k ends with discrepancy $n/2$; each new set has discrepancy at most L. Hence the greedy coloring still has discrepancy at least $n/2$, and the discrepancy of $\mathcal{R}$ increases by at most $\lceil \log_2 m \rceil$ compared to the original construction.

2. Determine the discrepancy of the complete set system that contains all subsets of X.

   Solution. For any coloring, there is a set in the complete set system that contains all the elements of the same color. Thus it is $\lceil n/2 \rceil$.

3. A graph G can be considered as a special set system in which all sets (edges) have size two. What can you say about the discrepancy of G?

   Solution. If there is no odd-length cycle in G, then G is bipartite. Such a graph has discrepancy 0, since we can color the two parts with the two colors. Otherwise, there is an odd cycle, and so there will be an edge with two endpoints of the same color, giving discrepancy of 2.

4. By using and tracking the weight function $y_S = 2^{\mathrm{disc}(S)}$, give a different proof of Theorem 1.2 (page 7).

Solution. At the start, the total weight is $2k \cdot 2^0 = 2k$. A direct induction gives the exact total weight at each iteration: after v_{2t} it is $2^{t-1}(4k - 2t + 1)$, and after v_{2t+1} it is $3 \cdot 2^{t-1}(2k - t)$. In particular, at the end, the total weight is $2^{k-1}(2k + 1)$.

By the pigeonhole principle, at the end, one set must have weight at least $2^{k-1}(2k+1)/(2k) = 2^{k-1}\left(1 + \frac{1}{2k}\right) > 2^{k-1}$. Taking logarithms, its discrepancy is at least $k = n/2$.

5. Give a complete proof of Lemma 1.1 (page 5).

   Hint: You may find the following version of Chernoff's bound useful:

   Lemma 1.3. *[Chernoff's bound] Let $X_1, \ldots, X_s$ be independent random variables with $X_i \in \{-1, 1\}$ and $\Pr[X_i = -1] = \Pr[X_i = 1] = 1/2$, for $i = 1, \ldots s$. Set $X = \sum_{i=1}^{s} X_i$. Then, for any $\Delta > 0$, we have*

   $$\Pr[|X| \geq \Delta] \leq 2e^{-\Delta^2/2s}.$$

   Solution. First, we consider the situation for a single set $S \in \mathcal{R}$. A simple calculation shows that *in expectation*, the random coloring χ of S is perfectly balanced:

   $$\mathrm{E}_\chi\left[\sum_{v \in S} \chi(v)\right] = \sum_{v \in S} \mathrm{E}_\chi\left[\chi(v)\right] = \sum_{v \in S}\left(\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot (-1)\right) = 0.$$

   However, we need the fact that χ is not only balanced in expectation, but *with high probability*, simultaneously for all sets. For this, we need a *concentration bound* which shows that the value of a random variable is close to its expected value, with high probability. In our setting, the most applicable concentration bound is *Chernoff's bound*. Using the version of Chernoff's bound that is given in the hint, we determine a $\Delta > 0$ such that the probability

   $$\Pr_\chi\left[\mathrm{disc}_\chi(S) \geq \Delta\right] = \Pr_\chi\left[\left|\sum_{v \in S} \chi(v)\right| \geq \Delta\right]$$

   is small. For this, let $S = \{v_1, \ldots, v_s\}$, $s = |S|$, and set $X_i = \chi(v_i)$, for $i = 1, \ldots, s$. Then, we have $X_i \in \{-1, 1\}$, and $\Pr[X_i = -1] = \Pr[X_i = 1] = 1/2$, independently of the others. We set $\Delta = \sqrt{2s \ln(2m^2)}$. Then, using Lemma 1.3 (page 12), we have

   $$\Pr_\chi\left[\left|\sum_{v \in S} \chi(v)\right| \geq \sqrt{2s\ln(2m^2)}\right] = \Pr_\chi\left[\left|\sum_{i=1}^{s} X_i\right| \geq \sqrt{2s\ln(2m^2)}\right]$$
   $$\leq 2e^{-(2s\ln(2m^2))/2s} = 2e^{-\ln(2m^2)} = \frac{1}{m^2}.$$

   Now, having analyzed the situation for a single set, let us bound the probability that there *exists* a set with discrepancy at least $\sqrt{2n\ln(2m^2)}$:

   $$\Pr_\chi\left[\text{there exists an } S \in \mathcal{R} \text{ with } \mathrm{disc}_\chi(S) \geq \sqrt{2n\ln(2m^2)}\right] \leq \sum_{S \in \mathcal{R}} \Pr_\chi\left[\mathrm{disc}_\chi(S) \geq \sqrt{2n\ln(2m^2)}\right]$$

   $$\leq \sum_{S \in \mathcal{R}} \Pr_\chi\left[\mathrm{disc}_\chi(S) \geq \sqrt{2|S|\ln(2m^2)}\right] \leq \sum_{S \in \mathcal{R}} \frac{1}{m^2} = \frac{1}{m}.$$

   Thus, we have

   $$\Pr_\chi\left[\text{for all } S \in \mathcal{R}, \text{ we have } \mathrm{disc}_\chi(S) < \sqrt{2n\ln(2m^2)}\right] \geq 1 - \frac{1}{m}.$$

   The claim follows, noting that $\sqrt{2n\ln(2m^2)} = O\left(\sqrt{n \ln m}\right)$.

6. Recall Lemma 1.3 (page 12):

Lemma 1.3. *[Chernoff's bound] Let $X_1, \ldots, X_s$ be independent random variables with $X_i \in \{-1, 1\}$ and $\Pr[X_i = -1] = \Pr[X_i = 1] = 1/2$, for $i = 1, \ldots s$. Set $X = \sum_{i=1}^{s} X_i$. Then, for any $\Delta > 0$, we have*

$$\Pr[|X| \geq \Delta] \leq 2e^{-\Delta^2/2s}.$$

Give a proof. Note that by symmetry, it suffices to show that

$$\Pr[X \geq \Delta] \leq e^{-\Delta^2/2s}.$$

Hint: Apply Markov's inequality to estimate the probability $\Pr[e^{\lambda X} \geq e^{\lambda\Delta}]$, for a parameter $\lambda > 0$, and use Fact A.22 (page 764). Then, choose an optimal value of λ.

Solution. ([Cha00, Lemma A.5])

$$\begin{aligned}
\Pr[X \geq \Delta] &= \Pr\left[e^{\lambda X} \geq e^{\lambda\Delta}\right] \leq e^{-\lambda\Delta} \cdot \mathrm{E}\left[e^{\lambda X}\right] \qquad \left(\text{Markov's inequality}\right) \\
&= e^{-\lambda\Delta} \cdot \mathrm{E}\left[\prod_{i=1}^{s} e^{\lambda X_i}\right] = e^{-\lambda\Delta} \cdot \prod_{i=1}^{s} \mathrm{E}\left[e^{\lambda X_i}\right] \\
&= e^{-\lambda\Delta} \cdot \prod_{i=1}^{s} \frac{1}{2}\left(e^{-\lambda} + e^{\lambda}\right) \leq e^{-\lambda\Delta} \cdot \prod_{i=1}^{s} \mathrm{e}^{\lambda^2/2} \qquad \left(\text{by Fact A.22 (page 764)}\right) \\
&= e^{-\lambda\Delta + \lambda^2 s/2} \leq e^{-\Delta^2/2s},
\end{aligned}$$

setting $\lambda = \Delta/s > 0$.

Now, Lemma 1.3 (page 12) follows by symmetry.

7. Consider the following online interval discrepancy game on a set system $(X, \mathcal{R})$, where $X = [n]$ and $\mathcal{R}$ consists of subsets induced by intervals on $[n]$:

   At iteration $t \in [n]$, the adversary gives a new point in $[n]$.
   Your goal is to color it immediately with a -1 or a $+1$ color.

   Give a strategy for the adversary such that no matter how you color the points of X, there will be an interval with discrepancy $\Omega\left(\sqrt{n}\right)$ at some point during the game.

   Solution. ([(ht]) Set $c \in (1/\sqrt{n}, 1/4]$ to be a fixed constant, say $c = 1/10$.

   Let S_j be the (signed) sum of all the colored elements after j iterations. If at any iteration this sum becomes greater than $c\sqrt{n}$ or less than $-c\sqrt{n}$, then clearly the interval containing $[n]$ gives the required lower bound. So assume that $S_j \in [-c\sqrt{n}, +c\sqrt{n})$ for all $j \in [n]$.

   As the value of the sum changes by $+1$ or -1 at each iteration, by the pigeonhole principle, there exists a value s such that $S_j = s$ and $S_{j+1} = s + 1$ for $r \geq \frac{n}{4c\sqrt{n}} - 1/2 = \sqrt{n}/4c - 1/2$ iterations $j \in [n]$ (the other case, that $S_{j+1} = s - 1$, is similar).

   In particular, the signed sum of all the elements of $[n]$ colored *during those iterations where the total sum is at most s*, must be at least

   $$r - 2c\sqrt{n} \geq \sqrt{n}\left(\frac{1}{4c} - 2c\right) - 1/2 = \Omega\left(\sqrt{n}\right).$$

   We'd be done if we could ensure that for any s, all the elements of $[n]$ colored while the sum is at most s, are in some interval. That is easy: map each value of the sum s to a set of $\sqrt{n}/2c$ consecutive points in $[n]$, sequentially over the values of s. That is, at each iteration j, if $S_j = t$, the adversary gives a new point from the interval

   $$\Big[\left(t + c\sqrt{n}\right)\sqrt{n}/2c, \left(t + c\sqrt{n}\right)\sqrt{n}/2c + \sqrt{n}/2c\Big).$$

   If at any iteration an interval becomes full, then we can stop and do the above argument for the corresponding sum value to get an interval with $\Omega\left(\sqrt{n}\right)$ discrepancy.

8. For the previous question, give a strategy that achieves discrepancy $O(\sqrt{n})$ at all times.

   Solution. ([Ger+25]) Partition $[n]$ into $\sqrt{n}$ consecutive intervals of length $\sqrt{n}$. For each such canonical interval, as the new points arrive that belong to it, we will alternately color them with -1 and $+1$. So at all times, the signed sum of the points in the same canonical interval is -1, 0 or $+1$.

   For any other interval I, if I fully contains a canonical interval, that adds $\{-1, 0, +1\}$ to its discrepancy, for a total of $\sqrt{n}$ value. On the other hand, the (at most) two canonical intervals containing the two endpoints of I can add a further $2\sqrt{n}$ to its discrepancy. This gives a total discrepancy of $O(\sqrt{n})$.

9. We now consider a famous problem where the greedy algorithm actually does give a good solution.

   Let $V = \{v_1, \ldots, v_n\}$ be n elements, and $\mathcal{S} = \{S_1, \ldots, S_m\}$ be m sets. Our goal is to compute a small set $H \subseteq V$ that is a hitting set for $\mathcal{S}$. Let OPT be the size of a minimum hitting set for $\mathcal{S}$.

   Prove that the following algorithm returns a hitting set H of size at most $\text{OPT} \cdot \ln m$.

   Set Cover $\left(V = \{v_1, \ldots, v_n\}, \mathcal{S} = \{S_1, \ldots, S_m\}\right)$

   Set $\omega^1(S) = 1$ for all $S \in \mathcal{S}$.
   Set $H = \emptyset$.
   For iteration $t = 1, \ldots, T$:

   1. pick an element $v^t \in V$ with:

      $$v^t = \arg\max_{v \in V} \sum_{\substack{S \in \mathcal{S}: \\ v \in S}} \omega^t(S).$$

   2. add v^t to H, and set the weights of all sets of $\mathcal{S}$ hit by v^t to 0:

      $$\omega^{t+1}(S) = \begin{cases} 0 & \text{if } v^t \in S \\ \omega^t(S) & \text{otherwise.} \end{cases}$$

   Return H.

   Solution. Set $\Omega^t = \sum_{S \in \mathcal{S}} \omega^t(S)$.

   The key idea is that at each step, by the pigeonhole principle, there is an element of the optimal hitting set that hits sets of total weight at least

   $$\frac{\Omega^t}{\text{OPT}},$$

   and thus we have

   $$\Omega^{t+1} \leq \Omega^t - \frac{\Omega^t}{\text{OPT}} = \Omega^t \cdot \left(1 - \frac{1}{\text{OPT}}\right).$$

   After T steps, assume that H is not a hitting set. Then we have

   $$1 \leq \Omega^{T+1} \leq m \cdot \left(1 - \frac{1}{\text{OPT}}\right)^T.$$

   Simplifying,

   $$\ln 1 \leq \ln m - \frac{T}{\text{OPT}}.$$

   Thus we can conclude that

   $$|H| = T \leq \text{OPT} \cdot \ln m.$$

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1.2 Discrepancy, I (algorithm for $\sqrt{n \ln m}$ using polynomials)

What you risk reveals what you value.

Jeanette Winterson

Given a finite set system $(X, \mathcal{R})$, one can ask whether a greedy coloring approach—one that colors the vertices of X one by one and ignores all vertices not yet assigned a color—can *ever* yield a coloring with small discrepancy for $\mathcal{R}$.

For instance, is it possible to get the coloring guaranteed by the following statement that we saw earlier:

Lemma 1.1. *Let $(X, \mathcal{R})$ be a finite set system, and let $n = |X|$ and $m = |\mathcal{R}| \geq 2$. Let χ be a random two-coloring of X, where, independently, each $v \in X$ is assigned one of the two colors with probability $1/2$. Then, with probability at least $1 - 1/m$,*

$$\operatorname{disc}_\chi(\mathcal{R}) = O\left(\sqrt{n \ln m}\right).$$

In this section, we answer this question affirmatively by presenting such an algorithm.

Theorem 1.4. *Given a finite set system $(X, \mathcal{R})$ with $m = |\mathcal{R}|, n = |X|$ such that $m \geq n \geq 2$, there is a greedy algorithm that colors the elements of X in the order $v_1, \ldots, v_n$ such that the final coloring $\chi \colon X \to \{-1, +1\}$ satisfies*

$$\text{for all } S \in \mathcal{R}\colon \quad \operatorname{disc}_\chi(S) = O\left(\sqrt{n \ln m}\right).$$

Here

$$\operatorname{disc}_\chi(S) = \left|\sum_{v \in S} \chi(v)\right|.$$

Further, the algorithm assigns a color to v_t in iteration t using only the information on elements $v_1, \ldots, v_t$.

Remark: The statement of Theorem 1.4 (page 16) holds for *every* ordering of the vertices.

Overview of ideas. The algorithm will color the elements in the order $v_1, \ldots, v_n$. When coloring v_t at iteration t, the previous algorithm focused only on the set of maximum discrepancy,

$$S^t = \arg \max_{\substack{S \in \mathcal{R}: \\ v_t \in S}} \operatorname{disc}^t(S),$$

where $\operatorname{disc}^t(S)$ is the discrepancy of S at the start of iteration t (uncolored elements are considered to have color 0).

Ignoring the discrepancy of all the other sets is a mistake.

Remark: Indeed, considering the lower bound construction from the previous section, the problem is that in each iteration we decrease the discrepancy of a single set, *while at the same time increasing the discrepancy of* $m/2$ *other sets*.

In the language of machine learning, we would say that the previous greedy algorithm *overfits*: it focuses too much on one constraint, and does not take the other constraints (or the future!) into account.

A typical way to deal with overfitting in the context of machine learning is called *regularization*: we modify the objective function so that the algorithm does not move too much in one direction.

We will now see how one can implement this idea in the context of discrepancy minimization.

Remark: Probabilistic proofs give another perspective on the tension between focusing on a single constraint and optimizing all constraints at the same time. There, measure concentration is used to estimate the probability of failure for a *single* set, and then the union bound is used to argue that the upper bound holds for *all* sets simultaneously.

The new idea is to assign a color to v_t by taking into account the discrepancies of all sets. We implement this by

a) devising a weight function that takes into account the current discrepancies of all sets of $\mathcal{R}$, and then

b) choosing the color of v_t so that this weight function is minimized.

Naturally, a set with higher discrepancy should contribute more to this weight; then, minimizing the total weight will implicitly restrain the discrepancy of any set from becoming too high.

Remark: A natural choice would be to consider the *sum* of all set discrepancies:

$$\sum_{S \in \mathcal{R}} \operatorname{disc}^t(S).$$

However, this again does not work: the influence of sets with small discrepancy may become too large, and this can lead to one set with big discrepancy (see exercises).

We need to increase the influence of sets with large discrepancy, while still taking all the sets into account. A flexible way to explore this approach is to try a *polynomial* weight function, Ω^t, defined as follows:

$$\Omega^t = \sum_{S \in \mathcal{R}} \underbrace{\left(\operatorname{disc}^t(S)\right)^c}_{\text{weight of } S}.$$

Here the parameter $c \geq 1$ can be chosen optimally later.

Remark: If c is too large, the above sum is dominated by the set realizing the maximum current discrepancy, and we revert back to the algorithm minimizing the maximum set discrepancy (which fails).

On the other hand, if c is too low, then a set can get high discrepancy, since it can be 'compensated' in the sum by many other sets having near-zero discrepancy—so this fails too (see exercises).

Since Ω^{t+1} will be the total weight once we have colored v_t, our goal is to give v_t a color that will minimize it.

What do we want for the final weight Ω^{n+1}, to be able to derive Theorem 1.4 (page 16)?

> By Lemma 1.1 (page 5), we know that there is a coloring such that $\operatorname{disc}(S) = O\left(\sqrt{n \ln m}\right)$ for all $S \in \mathcal{R}$. Thus, *ideally*, we would like the final weight to be
>
> $$\Omega^{n+1} = \sum_{S \in \mathcal{R}} \Big(\operatorname{disc}(S)\Big)^c = O\left(m \cdot \left(\sqrt{n \ln m}\right)^c\right).$$
>
> *If* this was the case, we can upper bound the discrepancy of any set $S \in \mathcal{R}$ as
>
> $$\Big(\operatorname{disc}(S)\Big)^c \leq \boldsymbol{\Omega^{n+1}} = O\left(m \cdot \left(\sqrt{n \ln m}\right)^c\right) \quad \Longrightarrow \quad \operatorname{disc}(S) = O\left(m^{\frac{1}{c}} \cdot \sqrt{n \ln m}\right).$$
>
> To get rid of the factor $m^{1/c}$, we should set $c \geq \ln m$, and we would recover the bound of Lemma 1.1 (page 5).

This plan is almost correct, except some technical details which need to be fixed carefully. To upper bound the weight increase at each iteration, the proof uses the following insight.

> For each $S \in \mathcal{R}$ containing v_t and with $\operatorname{disc}^t(S) > 0$, if one value of v_t's color increases its discrepancy, then the other value *must* decrease it.

Thus, by an averaging argument, there must exist a color for v_t that does not increase the total weight by too much.

In particular, we will show that if the discrepancy of a set S at the start of an iteration is at least $(nc)^{1/2}$, then the contribution of S to the total weight, averaged over the two possible colors of the next element, increases by a multiplicative factor of at most $1 + \frac{2c}{n}$; this follows from a careful calculation of this average:

$$\frac{1}{2}\Big((\operatorname{disc}(S)+1)^c + (\operatorname{disc}(S)-1)^c\Big) \leq \operatorname{disc}(S)^c \left(1 + \frac{2c}{n}\right).$$

This implies that

$$\Omega^{t+1} \leq m\,(4nc)^{\frac{c}{2}} + \left(1 + \frac{2c}{n}\right)\Omega^t,$$

which over n iterations gives

$$\Omega^{n+1} \leq nm\,(4nc)^{\frac{c}{2}} \cdot \left(1 + \frac{2c}{n}\right)^n = O\left(m^2\ (4nc)^{\frac{c}{2}}\ e^{2c}\right),$$

since $m \geq n$. An upper bound on the discrepancy of any $S \in \mathcal{R}$ now follows:

$$\Big(\operatorname{disc}(S)\Big)^c \leq \boldsymbol{\Omega^{n+1}} = O\left(m^2\ (4nc)^{\frac{c}{2}}\ e^{2c}\right) \quad \Longrightarrow \quad \operatorname{disc}(S) = O\left(m^{\frac{2}{c}} \cdot (nc)^{\frac{1}{2}}\right).$$

Setting $c = \Theta(\ln m)$ completes the proof sketch.

> **Remark:** One can see the precise Goldilocks tradeoff on c in the above bound, failing if c is too low, or too high.

Remark: The actual proof is more involved, for two reasons: first, the multiplicative bound on the weight increase holds only *on average*, and not for each set individually. This is not a problem, since we are upper bounding the sum in any case.

Second, the discrepancy of a single set S may oscillate above and below $\sqrt{nc}$ during the algorithm, and each time the discrepancy of S goes above $\sqrt{nc}$, it adds "fresh" weight to the total weight that is used in the averaging argument. However, calculations will show that this will increase the final weight by a negligible factor.

Remark: We emphasize a remarkable property of this algorithm: it works in an *online* fashion, that is, it can decide the color of v_t *without knowing anything about the next elements* $v_{t+1}, \ldots, v_n$.

In the previous section, we used exactly this property in order to construct a lower bound via an adversary argument. In this section, however, it turns out that by a slight modification of the weight function, we can circumvent this adversary argument and make a choice that will be useful no matter what the future holds.

This is a powerful property that makes the MWU technique applicable in many fields. As we will see later, this property is particularly useful in the fields of machine learning and online optimization.



Proof of Theorem 1.4 (page 16). We assume that $m < e^{n/8}$. Otherwise, $\sqrt{n \ln m} \geq n/\sqrt{8}$, and since $\mathrm{disc}_\chi(S) \leq |S| \leq n$ holds for any coloring χ and any $S \in \mathcal{R}$, every coloring trivially satisfies the claimed bound.

Here is the new algorithm. The integer parameter $c \in [1, n/2]$ will be fixed optimally later.

Sum Discrepancy Algorithm $\left(X = \{v_1, \ldots, v_n\}, \mathcal{R}\right)$

For $t = 1, \ldots, n$:

1. Assign v_t a color that *minimizes* the increase in the total weight of all the sets.
   That is, give v_t a color that realizes the following minimum:

$$\min\left\{\sum_{S\in\mathcal{R}}\left(\mathrm{disc}^{t+1}\left(S \mid v_t \text{ is given color '+1'}\right)\right)^c,\quad \sum_{S\in\mathcal{R}}\left(\mathrm{disc}^{t+1}\left(S \mid v_t \text{ is given color '-1'}\right)\right)^c\right\}.$$

Here is the key technical lemma that bounds the total weight increase at each iteration.

Claim 1.5. *During the t-th iteration, we can assign a color to v_t such that*

$$\Omega^{t+1} = \sum_{S\in\mathcal{R}}\left(\mathrm{disc}^{t+1}(S)\right)^c \leq m\,(4nc)^{\frac{c}{2}} + \left(1 + \frac{2c}{n}\right)\Omega^t.$$

Here integer c satisfies $1 \leq c \leq n/2$.

Proof. Set $a = (nc)^{1/2} \geq 1$.

For the sets that do not contain v_t, the weight remains unchanged. Fix a set $S \in \mathcal{R}$ that contains

v_t. We consider two cases:

Case 1: $\operatorname{disc}^t(S) < a$. In this case, irrespective of the color given to v_t, we have

$$\operatorname{disc}^{t+1}(S) < a+1 \qquad \Longrightarrow \qquad \left(\operatorname{disc}^{t+1}(S)\right)^c < (a+1)^c \le (4nc)^{\frac{c}{2}}.$$

Case 2: $\operatorname{disc}^t(S) \ge a$. Here, one of the color assignments for v_t must increase its discrepancy by 1, and the other must decrease it by 1.

Thus the average weight of S after coloring v_t, over the two possible color choices, is then

$$\frac{1}{2}\left(\operatorname{disc}^t(S)+1\right)^c + \frac{1}{2}\left(\operatorname{disc}^t(S)-1\right)^c = \left(\operatorname{disc}^t(S)\right)^c \left(\frac{1}{2}\left(1+\frac{1}{\operatorname{disc}^t(S)}\right)^c + \frac{1}{2}\left(1-\frac{1}{\operatorname{disc}^t(S)}\right)^c\right)$$

$$= \left(\operatorname{disc}^t(S)\right)^c \left(\frac{1}{2}\left(\sum_{i=0}^{c}\binom{c}{i}\frac{1}{\left(\operatorname{disc}^t(S)\right)^i} + \sum_{i=0}^{c}\binom{c}{i}\frac{(-1)^i}{\left(\operatorname{disc}^t(S)\right)^i}\right)\right) \qquad \left(\text{binomial theorem}\right)$$

$$= \left(\operatorname{disc}^t(S)\right)^c \left(1+\sum_{i=1}^{\lfloor c/2\rfloor}\binom{c}{2i}\frac{1}{\left(\operatorname{disc}^t(S)\right)^{2i}}\right) \qquad \left(\text{simplifying}\right)$$

$$\le \left(\operatorname{disc}^t(S)\right)^c \left(1+\sum_{i=1}^{\lfloor c/2\rfloor}\left(\frac{c^2}{i\left(\operatorname{disc}^t(S)\right)^2}\right)^i\right) \qquad \left(\text{using } \binom{N}{K} \le \frac{N^K}{(K/2)^{K/2}} = \left(\frac{\sqrt{2}\,N}{\sqrt{K}}\right)^K\right)$$

$$\le \left(\operatorname{disc}^t(S)\right)^c \left(1+\sum_{i=1}^{\lfloor c/2\rfloor}\left(\frac{c}{n}\right)^i\right) \qquad \left(\text{using } \left(\operatorname{disc}^t(S)\right)^2 \ge a^2 = nc\right)$$

$$\le \left(\operatorname{disc}^t(S)\right)^c \left(1+\frac{c}{n}\cdot\sum_{i=0}^{\infty}\left(\frac{c}{n}\right)^i\right) \le \left(\operatorname{disc}^t(S)\right)^c \left(1+\frac{2c}{n}\right),$$

where in the last step we assume that c will be chosen such that $c/n \le 1/2$.

Putting the above two cases together, the average value of Ω^{t+1} over the two choices can be upper bounded by

$$\sum_{\substack{S\in\mathcal{R}:\\ \operatorname{disc}^t(S)<a}} (4nc)^{\frac{c}{2}} + \sum_{\substack{S\in\mathcal{R}:\\ \operatorname{disc}^t(S)\ge a}} \left(\operatorname{disc}^t(S)\right)^c \left(1+\frac{2c}{n}\right) \le m\,(4nc)^{\frac{c}{2}} + \left(1+\frac{2c}{n}\right)\Omega^t.$$

By the pigeonhole principle, there is a color for v_t such that Ω^{t+1} is at most the above average. This completes the proof of the claim.

□

Using $\Omega^1 = 0$, Claim 1.5 (page 19), and unrolling the recurrence, we get

$$\Omega^{n+1} \le m\,(4nc)^{\frac{c}{2}} + \left(1+\frac{2c}{n}\right)\Omega^n \le m\,(4nc)^{\frac{c}{2}} + \left(1+\frac{2c}{n}\right)\cdot\left(m\,(4nc)^{\frac{c}{2}} + \left(1+\frac{2c}{n}\right)\Omega^{n-1}\right)$$

$$\le \cdots \le \sum_{t=0}^{n-1}\left(1+\frac{2c}{n}\right)^t \cdot m\,(4nc)^{\frac{c}{2}} \le nm\left(1+\frac{2c}{n}\right)^n (4nc)^{\frac{c}{2}}.$$

Thus, at the end of the algorithm, we can compare the upper and lower bounds on the final weight

to get

$$\max_{S\in\mathcal{R}}\left(\mathrm{disc}^{n+1}(S)\right)^c \le \sum_{S\in\mathcal{R}}\left(\mathrm{disc}^{n+1}(S)\right)^c = \boldsymbol{\Omega^{n+1}} \le nm\left(1+\frac{2c}{n}\right)^n (4nc)^{\frac{c}{2}},$$

implying that

$$\max_{S\in\mathcal{R}}\mathrm{disc}^{n+1}(S) \le (nm)^{\frac{1}{c}}\left(1+\frac{2c}{n}\right)^{n/c}(4nc)^{\frac{1}{2}} = O\left(m^{\frac{2}{c}}(nc)^{1/2}\right),$$

using $n \le m$, and $\left(1+\frac{2c}{n}\right)^{n/c} \le e^2$.

Setting $c = 2\lceil \ln m\rceil$ gives the desired bound.

Finally, we verify that our initial assumption $m < e^{n/8}$, and that $n \ge 8$, gives that

$$c = 2\lceil \ln m\rceil \le 2\lceil n/8\rceil \le n/2,$$

as needed earlier. □

Remark: The last step uses the fact that $\left(1+\frac{2c}{n}\right)^{n/c} = O(1)$. One can get an elementary proof using the binomial theorem, as follows:

$$\begin{aligned}
\left(1+\frac{2c}{n}\right)^{n/c} &\le \left(1+\frac{2c}{n}\right)^{\lceil n/c\rceil} = \sum_{i=0}^{\lceil n/c\rceil}\binom{\lceil n/c\rceil}{i}\left(\frac{2c}{n}\right)^i && \left(\text{binomial theorem}\right)\\
&\le 1+\sum_{i=1}^{\lceil n/c\rceil}\left(\frac{\sqrt{2}\lceil n/c\rceil}{\sqrt{i}}\right)^i\left(\frac{2c}{n}\right)^i && \left(\text{using }\binom{N}{K}\le\left(\frac{\sqrt{2}\,N}{\sqrt{K}}\right)^K\right)\\
&\le 1+\sum_{i=1}^{\infty}\left(\frac{\sqrt{2}\cdot 2(n/c)\cdot 2c}{\sqrt{i}\cdot n}\right)^i && \left(\text{using } c\le n/2 \text{ and } \lceil x\rceil \le 2x, \text{ for } x\ge 1/2\right)\\
&\le 1+\sum_{i=1}^{\infty}\left(\frac{8}{\sqrt{i}}\right)^i = O(1).
\end{aligned}$$

Remark: Claim 1.5 (page 19) computed the average of disc^{t+1} for each set. We can rephrase this succinctly in probabilistic language.

Set the color of v_t to $+1$ or -1 with equal probability. Then, for any set $S \in \mathcal{R}$ containing v_t and with $\mathrm{disc}^t(S) > 0$,

$$\mathrm{E}\left[\left(\mathrm{disc}^{t+1}(S)\right)^c\right] = \frac{1}{2}\left(\mathrm{disc}^t(S)+1\right)^c + \frac{1}{2}\left(\mathrm{disc}^t(S)-1\right)^c.$$

Thus there exists a color choice for v_t such that

$$\begin{aligned}
\Omega^{t+1} &\le \mathrm{E}\left[\sum_{S\in\mathcal{R}}\left(\mathrm{disc}^{t+1}(S)\right)^c\right] = \sum_{S\in\mathcal{R}}\mathrm{E}\left[\left(\mathrm{disc}^{t+1}(S)\right)^c\right]\\
&= \sum_{\substack{S\in\mathcal{R}:\\ \mathrm{disc}^t(S)<a}}\mathrm{E}\left[\left(\mathrm{disc}^{t+1}(S)\right)^c\right] + \sum_{\substack{S\in\mathcal{R}:\\ \mathrm{disc}^t(S)\ge a}}\mathrm{E}\left[\left(\mathrm{disc}^{t+1}(S)\right)^c\right]
\end{aligned}$$

$$\leq \sum_{\substack{S\in\mathcal{R}:\\ \operatorname{disc}^t(S)<a}} (2a)^c + \sum_{\substack{S\in\mathcal{R}:\\ \operatorname{disc}^t(S)\geq a}} \left(\operatorname{disc}^t(S)\right)^c \left(1+\frac{2c}{n}\right) \leq m(4nc)^{c/2} + \left(1+\frac{2c}{n}\right)\Omega^t.$$

For a set S not containing v_t, the discrepancy does not change. We can then complete the proof as earlier.



Bibliography and discussion. The algorithm and its analysis was constructed by the authors for pedagogical purposes.

EXERCISES

EXERCISES STILL UNDER CONSTRUCTION

1. Give a lower bound that shows that ***Sum Discrepancy Algorithm*** with $c = 1$ may construct a coloring with discrepancy $\Omega(n)$.

   Solution. We use an adversary argument as in Section 1.1 (page 7). We fix $k \in \mathbb{N}$, and we set $n = 2k + 1$. Our set system consists of just three sets A_1, A_2, and B. We assign the vertices $v_1, \ldots, v_n$ to the sets, in an online manner.

   On the arrival of v_1: add v_1 into A_1 and A_2. The sum discrepancy algorithm sets $\chi(v_1) = +1$, since all sets containing v_1 have discrepancy 0.

   On the arrival of v_2: add v_2 into A_1, A_2, and B. The sum discrepancy algorithm sets $\chi(v_2) = -1$, since this yields a total discrepancy of 1 (otherwise, we would get total discrepancy 5).

   On the arrival of v_3: add v_3 into A_1 and A_2. The sum discrepancy algorithm sets $\chi(v_3) = +1$, (both choices for $\chi(v_3)$ yield total discrepancy 3, so we may assume that the sum discrepancy algorithm uses the color $+1$).

   On the arrival of v_4: add v_4 into A_1, A_2, and B. The sum discrepancy algorithm sets $\chi(v_4) = -1$, since this yields a total discrepancy of 2 (otherwise, we would get total discrepancy 4).

   Situation before iteration $2t + 1$: We have $\operatorname{disc}^{2t+1}(A_1) = \operatorname{disc}^{2t+1}(A_2) = 0$, and $\operatorname{disc}^{2t+1}(B) = t$.

   On the arrival of v_{2t+1}: add v_{2t+1} into A_1 and A_2. The sum discrepancy algorithm sets $\chi(v_{2t+1}) = +1$, (both choices lead to a discrepancy of $t + 2$, so that the sum discrepancy algorithm uses the color $+1$).

   On the arrival of v_{2t+2}: add v_{2t+2} into A_1, A_2, and B. The sum discrepancy algorithm sets $\chi(v_{2t+2}) = -1$, since this yields a total discrepancy of $t + 1$ (otherwise, we would get total discrepancy $t + 3$.)

   After iteration $2k$, the set B has discrepancy $k = \lfloor n/2 \rfloor$. In the end, we add $v_n = v_{2k+1}$ to only A_1, so that the sets A_1 and A_2 are different. This does not affect the discrepancy of B.

2. Derandomize the proof of Lemma 1.1 (page 5) using the method of conditional expectations to get a different algorithm, that runs in deterministic time $O(nm)$.

   Solution. ([Cha00, Chapter 1.1]). We will construct a two-coloring χ with discrepancy at most ζ, where

   $$\zeta = \sqrt{4n \ln 2m}.$$

To derive an algorithm, it is useful to change our perspective slightly in the proof of Lemma 1.1 (page 5) (see Exercise 1.1.5 (page 12)): instead of working with probabilities, we switch to expectations. For each $S \in \mathcal{R}$, define the indicator random variable

$$X_S = \begin{cases} 1 & \text{if } \chi(S) \geq \zeta, \\ 0 & \text{otherwise.} \end{cases}$$

Let $X = \sum_{S \in \mathcal{R}} X_S$ be the number of sets of $\mathcal{R}$ with discrepancy at least ζ. Then

$$\mathrm{E}[X] = \mathrm{E}\left[\sum_{S \in \mathcal{R}} X_S\right] = \sum_{S \in \mathcal{R}} \mathrm{E}[X_S] = \sum_{S \in \mathcal{R}} \Pr[X_S = 1] \leq |\mathcal{R}| \cdot \frac{1}{2m^2} = \frac{1}{2m} < 1.$$

As $\mathrm{E}[X]$ is less than 1, there must exist a two-coloring for which $X < 1$, i.e., $X = 0$ as X is a non-negative integer. The key insight is the following:

$$\mathrm{E}[X] = \frac{1}{2}\mathrm{E}\left[X \mid \boldsymbol{\chi(v_1) = +1}\right] + \frac{1}{2}\mathrm{E}\left[X \mid \boldsymbol{\chi(v_1) = -1}\right].$$

Thus one of the two expectations on the R.H.S. must be at most $\mathrm{E}[X]$. Assuming it is the first one, we have

$$\mathrm{E}\left[X \mid \chi(v_1) = +1\right] \leq \mathrm{E}[X] < 1.$$

Thus we can set $\chi(v_1) = 1$, as there exists a setting of the remaining variables for which $X < 1$. We now repeat the same idea to fix the color of v_2:

$$\begin{aligned} \mathrm{E}\left[X \mid \chi(v_1) = +1\right] = & \frac{1}{2}\mathrm{E}\left[X \mid \chi(v_1) = +1,\ \boldsymbol{\chi(v_2) = +1}\right] \\ & + \frac{1}{2}\mathrm{E}\left[X \mid \chi(v_1) = +1,\ \boldsymbol{\chi(v_2) = -1}\right]. \end{aligned}$$

Continuing to fix the colors of $v_3, \ldots, v_n$ in this manner, we arrive at a coloring with $X < 1$; that is, all $S \in \mathcal{R}$ have discrepancy at most ζ.

Assume we have already fixed the color of $v_1, \ldots, v_{k-1}$; to set the color of v_k at the k-th step, we calculate the quantity

$$\begin{aligned} \mathrm{E}\left[X \mid \chi(v_1), \ldots, \chi(v_k) \text{ fixed}\right] &= \sum_{S \in \mathcal{R}} \mathrm{E}\left[X_S \mid \chi(v_1), \ldots, \chi(v_k) \text{ fixed}\right] \\ &= \sum_{S \in \mathcal{R}} \Pr\left[\chi(S) \geq \zeta \mid \chi(v_1), \ldots, \chi(v_k) \text{ fixed}\right]. \end{aligned}$$

Thus we need to compute, for every $S \in \mathcal{R}$ and iteration k, the expression

$$\Pr\left[\chi(S) \geq \zeta \mid \chi(v_1), \ldots, \chi(v_k) \text{ fixed}\right],$$

where the above probability is over the random colors of elements $v_{k+1}, \ldots, v_n$. Let $S' = S \setminus \{v_1, \ldots, v_k\}$ be the uncolored elements, and $t = \sum_{v \in S \cap \{v_1, \ldots, v_k\}} \chi(v)$ the sum of the colors already fixed. Then the above probability is simply

$$\Pr\left[\left|t + \sum_{v \in S'} \chi(v)\right| \geq \zeta\right],$$

or equivalently,

$$\Pr\left[\sum_{v \in S'} \chi(v) \leq -\zeta - t\right] + \Pr\left[\sum_{v \in S'} \chi(v) \geq \zeta - t\right].$$

The first event happens if at most $\frac{-\zeta - t + |S'|}{2}$ elements are colored '$+1$' in S', and the second happens if at least $\frac{\zeta - t + |S'|}{2}$ elements are colored '$+1$' in S'. This is precisely

$$\sum_{j=0}^{\lfloor \frac{-\zeta - t + |S'|}{2} \rfloor} \binom{|S'|}{j} \frac{1}{2^j} \frac{1}{2^{|S'|-j}} + \sum_{j=\lceil \frac{\zeta - t + |S'|}{2} \rceil}^{|S'|} \binom{|S'|}{j} \frac{1}{2^j} \frac{1}{2^{|S'|-j}}$$

$$= \frac{1}{2^{|S'|}} \left(\sum_{j=0}^{\lfloor \frac{-\zeta - t + |S'|}{2} \rfloor} \binom{|S'|}{j} + \sum_{j=\lceil \frac{\zeta - t + |S'|}{2} \rceil}^{|S'|} \binom{|S'|}{j} \right).$$

One can pre-compute all summations above at the start, so that at each iteration, for each $S \in \mathcal{R}$, the above probability can be computed in $O(1)$ time. Each iteration requires computing m such terms, and there are n iterations in total. Thus the overall time to compute χ is $O(nm)$.

3. Let X be a non-negative random variable. For a parameter $\delta > 0$, we wish to upper bound

$$\Pr[X \geq \delta].$$

The way to amplify this bound is by applying Markov's inequality on a transformed quantity. Two such transformations are:

Moment generating function: $\Pr\left[e^{\lambda X} \geq e^{\lambda \delta}\right] \leq \dfrac{\mathrm{E}\left[e^{\lambda X}\right]}{e^{\lambda \delta}}$. (Exercise 1.1.6 (page 12))

Polynomial function: $\Pr\left[X^k \geq \delta^k\right] \leq \dfrac{\mathrm{E}\left[X^k\right]}{\delta^k}$.

Prove that there is always a choice of k for which the polynomial function is no worse than the moment generating function. That is, for any fixed λ, δ, we have

$$\inf_{k \geq 0} \frac{\mathrm{E}\left[X^k\right]}{\delta^k} \leq \frac{\mathrm{E}\left[e^{\lambda X}\right]}{e^{\lambda \delta}}.$$

Solution. ([PN95]; see also [Wai19])

$$\mathrm{E}\left[e^{\lambda X}\right] = \mathrm{E}\left[\sum_{i=0}^{\infty} \frac{(\lambda X)^i}{i!}\right] = \sum_{i=0}^{\infty} \frac{(\lambda \delta)^i}{i!} \cdot \mathrm{E}\left[\frac{X^i}{\delta^i}\right]$$

$$\geq \sum_{i=0}^{\infty} \frac{(\lambda \delta)^i}{i!} \cdot \inf_{j \geq 0} \mathrm{E}\left[\frac{X^j}{\delta^j}\right] = \left(\inf_{j \geq 0} \mathrm{E}\left[\frac{X^j}{\delta^j}\right]\right) \sum_{i=0}^{\infty} \frac{(\lambda \delta)^i}{i!}$$

$$= \left(\inf_{j \geq 0} \frac{\mathrm{E}\left[X^j\right]}{\delta^j}\right) e^{\lambda \delta}.$$

4. Let $X = \sum_{i=1}^{n} X_i$, where, independently, each X_i is set to $+1$ or -1 with probability $1/2$. Prove that for a positive integer c,

$$\mathrm{E}[X^c] = \sum_{\substack{c_1, c_2, \ldots, c_n \in \mathbb{N}_0 \\ c_1 + c_2 + \cdots + c_n = c \\ \text{all } c_i\text{'s even}}} \binom{c}{c_1, c_2, \ldots, c_n}.$$

Deduce that $\mathrm{E}[X^c] \leq (\beta n c)^{c/2}$, for some fixed constant $\beta > 0$ that is independent of n and c. Then use linearity of expectation to prove the existence of a two-coloring, for a set system with n vertices and m sets, with discrepancy $O\left(\sqrt{n \ln m}\right)$.

Solution.

$$\mathrm{E}[X^c] = \mathrm{E}\left[\left(\sum_{i=1}^{n} X_i\right)^c\right]$$

$$= \mathrm{E}\left[\sum_{\substack{c_1,c_2,\ldots,c_n\in\mathbb{N}_0\\ c_1+c_2+\cdots+c_n=c}} \binom{c}{c_1,c_2,\ldots,c_n} \prod_{i=1}^{n} X_i^{c_i}\right] \quad \text{(multinomial theorem)}$$

$$= \sum_{\substack{c_1,c_2,\ldots,c_n\in\mathbb{N}_0\\ c_1+c_2+\cdots+c_n=c}} \binom{c}{c_1,c_2,\ldots,c_n} \prod_{i=1}^{n} \mathrm{E}[X_i^{c_i}], \quad \text{(linearity of expectation, independence)}$$

where

$$\binom{c}{c_1,c_2,\ldots,c_n} = \frac{c!}{c_1!c_2!\cdot\cdots\cdot c_n!}$$

is the multinomial coefficient, which counts the ways how c elements can be partitioned into n distinct sets such that set i has c_i elements.

For each $i \in [n]$, by definition of expectation,

$$\mathrm{E}[X_i^{c_i}] = \frac{1}{2}\cdot(-1)^{c_i} + \frac{1}{2}\cdot 1^{c_i} = \begin{cases} 0, & \text{if } c_i \text{ is odd,} \\ 1, & \text{if } c_i \text{ is even.} \end{cases}$$

Thus, it follows that

$$\mathrm{E}[X^c] = \sum_{\substack{c_1,c_2,\ldots,c_n\in\mathbb{N}_0\\ c_1+c_2+\cdots+c_n=c\\ \text{all } c_i\text{'s even}}} \binom{c}{c_1,c_2,\ldots,c_n}.$$

Note that $\mathrm{E}[X^c] = 0$ if c is odd.

So assume that c is even. Since $\sum_{i=1}^{n} c_i = c$ and since all c_i's are even (in particular, no c_i is 1), we can conclude that at most $c/2$ of the c_i's are non-zero. Fix a parameter $\ell \in \{0,\ldots,c/2\}$ such that exactly ℓ indices c_i are non-zero, and we bound the sum of multinomial coefficients for indices that fulfill this additional constraint. For this, we first note that under these conditions, we have

$$\binom{c}{c_1,c_2,\ldots,c_n} \le \ell^c,$$

because the left-hand side counts the number of ways a set of c elements can be partitioned into ℓ distinguishable sets of given sizes $c_1,\ldots,c_n$ (where only ℓ of the c_i's are non-zero), while the right-hand counts the number of ways a set of c elements can be partitioned into ℓ distinguishable sets of arbitrary sizes. There are $\binom{n}{\ell}$ ways to choose the c_i's that are non-zero, so we get

$$\sum_{\substack{c_1,c_2,\ldots,c_n\ge 0\\ c_1+c_2+\cdots+c_n=c\\ \text{all } c_i\text{'s even, exactly } \ell \text{ of the } c_i\text{'s non-zero}}} \binom{c}{c_1,c_2,\ldots,c_n} \le \binom{n}{\ell}\cdot\ell^c \le \left(\frac{en}{\ell}\right)^{\ell}\cdot\ell^c,$$

using the well-known estimate $\binom{n}{\ell} \le (\frac{en}{\ell})^{\ell}$.

Thus, from the discussion so far, we get that

$$\mathrm{E}[X^c] \le \sum_{\ell=0}^{c/2} \left(\frac{en}{\ell}\right)^{\ell}\cdot\ell^c.$$

This is an increasing geometric series: if $c \le n$, we have that

$$\frac{\left(\frac{en}{\ell+1}\right)^{\ell+1}\cdot(\ell+1)^c.}{\left(\frac{en}{\ell}\right)^{\ell}\cdot\ell^c.} = en\cdot\frac{\ell^\ell}{(\ell+1)^{\ell+1}}\cdot\frac{(\ell+1)^c}{\ell^c} \ge 2,$$

and the sum is asymptotically dominated by the term for $\ell = c/2$. We obtain

$$\mathrm{E}[X^c] \leq (\beta nc)^{c/2},$$

for some fixed constant $\beta > 0$ that is independent of n and c, and this also holds for $c > n$, since we always have $X^c \leq n^c$.

To prove the discrepancy bound, for any $S \in \mathcal{R}$ we have

$$\mathrm{E}\left[\mathrm{disc}(S)^c\right] = O\left(|S|c\right)^{c/2} = O\left(nc\right)^{c/2} \quad \Longrightarrow \quad \mathrm{E}\left[\sum_{S\in\mathcal{R}} \mathrm{disc}(S)^c\right] = m \cdot O\left(nc\right)^{c/2}.$$

Thus, there exists a coloring of X with

$$\sum_{\boldsymbol{S\in\mathcal{R}}} \mathbf{disc}(\boldsymbol{S})^{\mathbf{c}} \leq m \cdot O\left(nc\right)^{c/2}.$$

It follows that for every $S \in \mathcal{R}$, we have

$$\mathrm{disc}(S)^c \leq m \cdot O\left(nc\right)^{c/2} \qquad \Longrightarrow \qquad \mathrm{disc}(S) = m^{1/c} \cdot O(\sqrt{nc}).$$

Setting $c = 2\lceil \ln m \rceil$ proves the bound.

5. Let $X = \sum_{i=1}^{n} X_i$, where each X_i is set to $+1$ or -1 independently with probability $1/2$. Then use Markov's inequality together with the bound in Exercise 1.2.4 (page 24) on $\mathrm{E}\left[X^c\right]$ to prove that

$$\Pr[|X| \geq \alpha\sqrt{n}] \leq e^{-c'\alpha^2},$$

where $c' > 0$ is a constant independent of n and α. Use this to prove Lemma 1.1 (page 5).

Solution. We use Markov's inequality to obtain the following deviation bound for X, for any even $c \geq 0$ (we need c to be even, because Markov's inequality applies only to nonnegative random variables):

$$\Pr[|X| \geq \alpha\sqrt{n}] = \Pr[X^c \geq \alpha^c n^{c/2}] \leq \frac{\mathrm{E}[X^c]}{\alpha^c n^{c/2}} \leq \frac{(\beta nc)^{c/2}}{\alpha^c n^{c/2}} = \left(\frac{\beta c}{\alpha^2}\right)^{c/2}.$$

Now, let us fix an $\alpha > 0$, and let us determine the exponent c for which this upper bound is minimized. Taking the derivative, we get

$$\frac{\partial}{\partial_c}\left(\frac{\beta c}{\alpha^2}\right)^{c/2} = \frac{\partial}{\partial_c}\exp\left(\frac{c}{2}\cdot\ln\left(\frac{\beta c}{\alpha^2}\right)\right) = \left(\frac{1}{2}\cdot\ln\left(\frac{\beta c}{\alpha^2}\right) + \frac{c}{2}\cdot\frac{\beta}{\alpha^2}\cdot\frac{\alpha^2}{\beta c}\right)\cdot\exp\left(\frac{c}{2}\cdot\ln\left(\frac{\beta c}{\alpha^2}\right)\right)$$

$$= \left(\frac{1}{2}\cdot\ln\left(\frac{\beta c}{\alpha^2}\right) + \frac{1}{2}\right)\cdot\left(\frac{\beta c}{\alpha^2}\right)^{c/2}.$$

Setting the derivative to zero, we get

$$\frac{1}{2}\cdot\ln\left(\frac{\beta c}{\alpha^2}\right) + \frac{1}{2} = 0, \qquad \Longleftrightarrow \qquad \ln\left(\frac{\beta c}{\alpha^2}\right) = -1.$$

Thus setting $c = \frac{\alpha^2}{e\beta}$, rounding to make it an even integer,

$$\Pr[|X| \geq \alpha\sqrt{n}] \leq \left(\frac{\beta c}{\alpha^2}\right)^{c/2} = e^{-c'\alpha^2},$$

where $c' > 0$ is a constant independent of n and α, and α is a large-enough constant.

Proof of Lemma 1.1 (page 5) follows, as the probability that the discrepancy of any $S \in \mathcal{R}$ is greater than $\sqrt{2\frac{n\ln m}{c'}}$ is less than $1/m^2$, and union bound completes the proof.

[Cha00] B. Chazelle. *The Discrepancy Method: Randomness and Complexity*. Cambridge University Press, 2000 (cit. on pp. 11, 13, 22, 40, 50, 261, 507, 576).

[PN95] T. K. Philips and R. Nelson. “The Moment Bound Is Tighter Than Chernoff’s Bound for Positive Tail Probabilities”. In: *The American Statistician* 49.2 (1995), pp. 175–178 (cit. on p. 24).

[Wai19] M. J. Wainwright. *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*. Cambridge University Press, 2019 (cit. on p. 24).